

Estimating Road-Test Routes from Partial, Non-Independent Crowd Reports

Evidence aggregation and validation in OntarioDriveTestMap

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Abstract

Ontario does not publish the routes used in its G2 and G road tests. OntarioDriveTestMap reconstructs them from what test-takers leave behind in public: Reddit posts and dashcam video. Every such account is partial, some are copies of others, and none can be checked against an official answer. This report treats route reconstruction as a problem of *statistical inference under disagreement*. It describes a pipeline that turns posts and video frames into ordered street sequences, resolves consecutive streets to junctions on a local OpenStreetMap index, and scores each road segment by author-deduplicated, source-weighted, age-decayed support. The estimator is evaluated with leave-one-trace-out validation at three centres (Walkley, Canotek, Smiths Falls).

Four results stand out. First, the publish threshold fixed at design time was wrong everywhere it was tested: at $\tau = 1.5$ the segment-level F_1 is 0.64, 0.00 and 0.00, against 0.79, 0.56 and 0.91 at the measured optimum. Second, the weighting machinery did not beat a raw mention count. At Walkley both score F_1 0.79 (posterior difference -0.06 to $+0.07$). At Canotek the weighting was worse (0.56 against 0.77) until its age decay was relaxed and more real corroboration was added (0.78 against 0.79). Third, pooling distinct routes into one consensus cost 0.08 to 0.09 F_1 at Walkley when first measured, but that advantage did not survive later corpus growth. Fourth, cameras record streets they pass as well as streets they turn onto. This *drive-past* bias gave the highest-support segment in the corpus to a street the route passes but never turns onto, and it survives every filter that relies on video alone.

Observations are clustered within 8 to 27 pseudonymous authors per centre. Once that clustering is allowed for, intervals on every headline figure widen sharply, and differences below roughly 0.2 in F_1 cannot be resolved. Leave-one-out also measures agreement among sources, not correctness. The report closes with the hand-transcribed routes now shipped on the live map, and with how they could supply the external ground truth this analysis lacks.

1 Introduction

Road tests in Ontario are run at DriveTest centres, operated by Serco on behalf of the Ministry of Transportation (MTO). Examiners follow routes that are not published. Several commercial and free websites claim to show them. A survey of those sites during this project found none backed by observed data. The most prominent paid site states in its own disclaimer that its free sample route does not represent an actual test route. Another labels its map a schematic that is not to scale.

The information does exist, scattered across the public web. People write up their test on Reddit,

sometimes turn by turn. Driving instructors and test-takers upload dashcam recordings to YouTube. OntarioDriveTestMap collects these accounts and asks a statistical question of them:

Given many partial, noisy, possibly copied accounts of an unobserved route, with no official answer to check against, which road segments belong to the route, and how confident should a reader be in each one?

The problem has the shape of consensus assembly, where one sequence is reconstructed from many overlapping noisy reads. It also has four properties that rule out the textbook versions.

- (i) **Partial observation.** Most accounts cover a fraction of the route. A post that says “I went down Walkley, then got on the parkway” skips every street in between.
- (ii) **Non-independence.** One author may post a route, then relay three more in a reply. One channel may upload the same drive years apart. A post may retell a video.
- (iii) **Unknown reliability.** Nothing says in advance whether a given post is firsthand or remembered, current or years stale.
- (iv) **No ground truth.** The authority that knows the routes does not publish them, so accuracy can only be estimated indirectly.

Contributions. This report documents the method as built, and the evidence for and against each of its parts.

- A measurement pipeline from posts and video frames to junction-resolved street segments. It rests on a local intersection index derived from OpenStreetMap (OSM) in place of a geocoder (Section 3).
- A per-segment support estimator with author deduplication, source weighting and exponential age decay, plus a clustering step that groups traces into route families (Section 4).
- Leave-one-trace-out evaluation against frequency and random baselines at three centres, including a threshold sweep (Sections 5 and 6).
- Interval estimates that make the dependence between observations explicit, and a discussion of what the headline numbers do and do not support (Sections 7 and 8).
- An account of the *drive-past* bias, a failure mode specific to combining video and text, and the only signal found that separates it from genuine turns (Section 6.4).

Scope. The project launched with four centres: two high-volume Ottawa centres (Walkley and Canotek) and two low-volume rural ones (Smiths Falls and Winchester). Winchester is excluded from the analysis. Its 12 collected Reddit threads named no streets at all, and all 16 of its processed videos turned out to be recorded in Winchester, England (Section 2.3). Every number in this report was measured and logged when it was produced. Nothing was re-run to write it. Where a figure was later found to be wrong, the correction is reported with it (Appendix D).

Table 1: Corpus funnel by centre. Counts are as logged at collection. “Traceable” means a record with at least two ordered turns naming a street.

	Walkley	Canotek	Smiths Falls
Reddit records	1,259 (111 posts, 1,148 comments), 532 authors	408 (62 posts, 346 comments), 140 authors	58 unique threads, read in full
Traceable records / traces	14 at first, 18 after parser fixes (1.4%)	11 (2.7%)	2 at first; 7 traces by the end
YouTube candidates	112 raw, 94 kept	185 raw, 78 kept	291 raw, 70 kept
Usable route videos	~36 recordings from ~20 channels	12 verified; 7 yielded OCR traces	4 dashcam videos; 1 with usable text

2 Data

2.1 Sources and collection

Reddit. The plan was to search Reddit programmatically. That failed three times in two days. Public JSON endpoints returned HTTP 403. Creating an API app failed silently: the form returned HTTP 200 with a success flag, and the real message, a pointer to Reddit’s approval policy, sat in the response body. A data-access request was then rejected. Reddit threads were therefore saved by hand from r/Ontariodrivetest and r/ottawa and parsed offline. Author names were replaced at ingest by a salted SHA-256 pseudonym, so no username is stored. The pseudonym still lets the pipeline tell when two records come from the same person, which matters later (Section 7).

YouTube. Candidate videos were found through the YouTube Data API. The collector searches several query variants per centre, deduplicates by video ID and records metadata. Route evidence was then extracted from the video frames (Section 3.4). Captions were collected too, but they turned out not to be a usable source of route order (Section 2.2).

Table 1 summarises the funnel. Two things stand out. The first is how little of the Reddit text carries route geometry. Of the 1,259 Walkley records, 92.3% contain none; they are outcome reports, nerves, booking questions and manoeuvre advice. The second is that deliberate collection pays. Walkley’s Reddit corpus was gathered while searching broadly, while Canotek’s was collected for Canotek specifically. The traceable rate roughly doubled, from 1.4% to 2.7%.

2.2 The structural fact about the data

One pattern shaped every later design decision: *sources either name streets or give their order. Almost none do both.*

- **Reddit prose** names streets and skips the connections between them. The writer knows the route and compresses it.
- **Reddit route lists** (“Route 1: left on Walkley, right on Cedarwood. . .”) give both, but are rare. All four traces that resolved fully in the first snapping run were lists of this kind, and every prose account failed or came back partial. Format predicted usability better than platform did.
- **Transcripts** name streets without order. Of 94 Walkley videos, 17 returned a usable transcript (49 had captions disabled, 25 hit rate limits, 3 had none), and together they produced **zero** traces. Drivers narrate streets as description (“that’s the Airport Parkway up ahead”). The

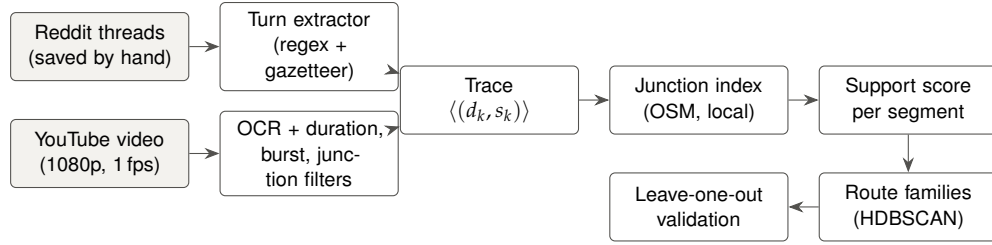


Figure 1: Pipeline. Both source types converge on one intermediate form, an ordered list of (direction, street) pairs. Everything downstream is source-agnostic.

instruction stays “turn left here”.

- **Video frames** give both. Every street sign read from a frame carries a timestamp, so order comes free.

This inverts the original design, which treated captions and transcripts as the main extraction path and optical character recognition (OCR) on frames as a fallback for videos with a GPS overlay. Measured, OCR is the main path and text is corroboration.

2.3 Collector metadata is a hypothesis, not a fact

Three metadata fields proved less reliable than the data they described.

- **Centre.** The first collector stamped the search term into a *centre* column. All 94 Walkley rows read “walkley”, including two videos titled Canotek and one titled Smiths Falls. A selection by class and duration picked three long videos from other parts of the city, and frames were extracted from all three before local knowledge caught it.
- **Test class.** Guessing class from titles gave 47 G, 7 G2, 21 ambiguous and 19 unknown for Walkley. Six of the first eight OCR traces carried “ambiguous” or “unknown”.
- **Place.** Every one of the 16 processed “Winchester” videos was recorded in Winchester, Hampshire. The OCR text named local roads, UK driving-school branding, a GPS readout at N51.0/W1.3 and the M3 to Southampton. Title-level filters caught 15 UK videos earlier, but for most of the rest the evidence was visible only in the footage itself.

The working principle that followed was to derive attributes from content wherever the content can support them. As Section 6.3 shows, test class can be recovered from geometry alone.

3 Measurement: from sources to segments

3.1 Representation

A *trace* t is an ordered list of turns $\langle (d_1, s_1), \dots, (d_m, s_m) \rangle$, each a direction and a normalised street name, together with a pseudonymous author $a(t)$, a source type and a date. A *segment* is an unordered pair of consecutive distinct streets $e = \{s_k, s_{k+1}\}$ that meet at a junction on the road graph. A pair that does not meet is dropped, and the reason is recorded. The unit of analysis is therefore the junction transition, not a polyline. Two recordings of the same turn produce the same segment even when their GPS paths would differ by metres.

The design called for sequences of OSM way IDs, obtained by map-matching each trace. Street-pair segments replaced them because they can be computed directly from the index described next,

and because every source, including a one-line Reddit comment, can produce them.

3.2 A local intersection index instead of a geocoder

The original plan geocoded intersections with the public Nominatim service. Nominatim has no concept of an intersection. Two of three test lookups succeeded only because OSM happened to contain named junction nodes, and the third failed on a junction that exists. Overpass returned timeouts and a mirror returned HTTP 500. The replacement exploits a fact already present in the map data: an intersection is a node shared by two or more named drivable ways. One pass over a 47 MB Ottawa extract produced:

Named drivable ways	36,534	Junction nodes	19,482
Nodes touched	219,761	Distinct street names	9,879

Every junction was located, and lookups became offline and exact. Two later changes followed from failures.

- **Ramps.** Grade-separated roads connect through unnamed link ways, so Airport Parkway, a multi-kilometre arterial, had only two junctions in the whole extract. Six independent authors describing airport parkway $\times$ walkley were being dropped as impossible. Unnamed link ways are now grouped into connected components by union-find, and every named street touching a component is treated as connected to every other. That added 835 ramp connections (+4%). The segment then became the best-supported one in the corpus at the time.
- **Coverage.** Checked directly, the extract turned out not to reach the rural centres; the design had assumed it did. It was rebuilt from a provincial extract, clipped to a box around all four centres. Smiths Falls went from 0 to 495 junctions and Winchester to 172. Eight high-weight Walkley segments were verified to have identical node IDs and coordinates in both databases before the swap.

3.3 Text extraction

Turns are extracted from prose with regular expressions and validated against every street name in the extract. The first pass produced about 40% junk. Writers run the street name straight into the sentence, so “right on Walkley you pass the...” was captured as a street called “walkley you pass”. The fix truncates at the first token that cannot belong to a street name, caps names at three tokens, unifies suffix variants, applies an alias table for common misspellings and fuzzy-matches against the gazetteer.

The cutoffs are deliberately conservative: 0.86 against a local list, then 0.88 against 17,302 names. The costs are asymmetric. A missed street loses one trace. A wrongly matched street enters the graph as a real edge and looks like consensus. Five parser bugs found while processing the second and third centres are now encoded, verbatim from the source text, as a regression suite, alongside four documented known gaps.

3.4 Video extraction

Frames are sampled at 1 fps and read with PaddleOCR, and each text box is matched against the gazetteer for streets near the centre. Settings were chosen by measurement on one 18-minute video (Table 2). Resolution was the dominant factor. Downscaling to 1280 px destroyed residential blade-sign text, and cropping to the top of the frame removed signs that sit lower than assumed.

Table 2: OCR configuration on one 18.3-minute Walkley G2 video. A *hit* is a frame naming a gazetteer street.

Configuration	Frames	Hits	Streets found
720p → 1280 px, top-65% crop, 0.5 fps	548	2	walkley
720p → 1280 px, no crop, 0.5 fps	548	7	walkley
1080p → 1920 px, no crop, 1 fps, fuzzy 0.85	1,095	114	walkley, heatherington, fairlea, baycrest

Four filters turn raw sightings into a trace. Each was added in response to a failure seen in real output.

Duration. Real signs were visible for 1–9 s and false positives for 18–43 s. Those were static text inside the car or a parking sign held in view. A cutoff near 15 s separated them perfectly on the calibration video. Unlike a word list, it does not need to anticipate what is inside a given driver’s car.

Burst. A frame naming four or more streets is a map overlay or a title card. Two “route guide” videos opened with eight streets in alphabetical order. Two others matched on every frame (2,029 of 2,033 and 1,401 of 1,403) because a map sat permanently on screen; the filters reduced them to 0 and 7 sightings.

Junction. Consecutive sightings must meet on the graph, within a window of ± 2 . This removed shopfronts and misreads.

Dedupe. Repeated readings of the same sign are collapsed.

A free-tier cloud GPU processed 15 videos (about 330 minutes of footage) in roughly five hours. A local CPU run measured about 20 s per frame, around $100\times$ slower. The earlier estimate of $3\times$ had compared runs at different resolutions.

3.5 Completeness differs sharply by source

Once snapped to the graph, the two source types behave very differently. Snapping gives each trace one of three statuses: *full* (every consecutive pair resolves to a junction), *partial*, or *failed*. **No video trace failed (0 of 20; 95% Wilson interval 0–16%), against 11 of 20 text traces (34–74%). Fisher’s exact test gives $p \approx 1.5 \times 10^{-4}$.** A camera passes every sign on the road, while a person recalling a route skips connecting streets. That motivated weighting sources by completeness rather than by platform. Section 6.4 shows that the same property also has a cost.

4 The estimator

4.1 Per-segment support

Let $\mathcal{T}$ be the traces for one centre, and let $\mathcal{A}(e)$ be the set of authors with at least one trace containing segment e . A YouTube channel counts as one author. Each trace has a weight $w(t) = w_{\text{src}}(t) w_{\text{age}}(t)$. The support of a segment is

$$S(e) = \sum_{a \in \mathcal{A}(e)} \max_{\substack{t: a(t)=a \\ e \in t}} w(t), \quad (1)$$

and e is predicted to lie on the route when $S(e) \geq \tau$. The inner maximum is the *author deduplication*: an author who wrote four traces counts once, at their strongest. Alongside $S(e)$ the pipeline reports the raw author count $|\mathcal{A}(e)|$. The count is a fact, while the weight embeds judgement, and the map needs to be able to say “four independent people” without defending a formula.

Source weight. Video traces receive $w_{\text{src}} = 0.9$. Reddit traces receive 0.25, 0.45 or 0.60, rising with the number of extracted turns. These values stand in for completeness (Section 3.5). They were never fitted.

Age weight. Routes change, so older evidence is discounted exponentially rather than cut off:

$$w_{\text{age}}(t) = \max(0.15, 2^{-\Delta_t/h}), \quad (2)$$

where Δ_t is the trace’s age in years and h is the half-life. Undated traces receive 0.6. The half-life was $h = 3$ years at design time and was changed to $h = 5$ in Section 6.2. The logged per-trace weights confirm Equation (2). For example, a 2021 trace scored 0.31 at $h = 3$, and $2^{-5.03/3} = 0.313$.

4.2 Baselines

Two baselines use the same leave-one-out harness. The **frequency** baseline scores a segment by how many times it is mentioned, with no author deduplication, no source weights and no age decay. The **random** baseline gives a floor. If the weighted estimator cannot beat raw frequency, its three mechanisms are adding complexity without adding accuracy.

4.3 Route families

A centre runs several routes per test class, and they share streets near the centre. Traces are therefore grouped into *families* before a consensus is built. Each trace is represented as its set of segments. Similarity is a weighted overlap coefficient,

$$\text{sim}(A, B) = \frac{W(A \cap B)}{\min\{W(A), W(B)\}}, \quad W(X) = \sum_{e \in X} \text{idf}(e), \quad \text{idf}(e) = \ln \frac{N}{\text{df}(e)}, \quad (3)$$

where $\text{df}(e)$ counts the traces containing e . The logged IDF values match this form: on 17 traces, a segment present in 11 scored 0.44 ($\ln \frac{17}{11} = 0.435$) and one present in a single trace scored 2.83 ($\ln 17$). The overlap coefficient was chosen over Jaccard because traces are partial, and Jaccard would score containment as dissimilarity. The IDF weighting was added after the data showed a shared spine dominating similarity: almost every Walkley trace runs along Walkley Road, so plain overlap rated any two traces as alike. Families are found with HDBSCAN on $1 - \text{sim}$, with a minimum family size of three. HDBSCAN suits the problem for three reasons: the number of routes is unknown, a set of segments has no meaningful centroid, and points labelled as noise flag unreliable traces at no extra cost.

4.4 Why the whole subgraph is emitted, not a single best path

The design specified Dijkstra’s algorithm over $-\log(\text{support}/\text{total})$ to extract one highest-support path per family. It was not built. Walkley G2 is one arterial spine with two residential loops hanging off it. A single best path would keep the better-supported loop and silently drop the other. Instead, every above-threshold segment is emitted and split into runs where the graph branches. Branching is information the reader should see.

5 Evaluation design

Leave-one-trace-out. For each trace t_i , the consensus is rebuilt from $\mathcal{T} \setminus \{t_i\}$ and its predicted segments are compared with the segments of t_i . True positives, false positives and false negatives

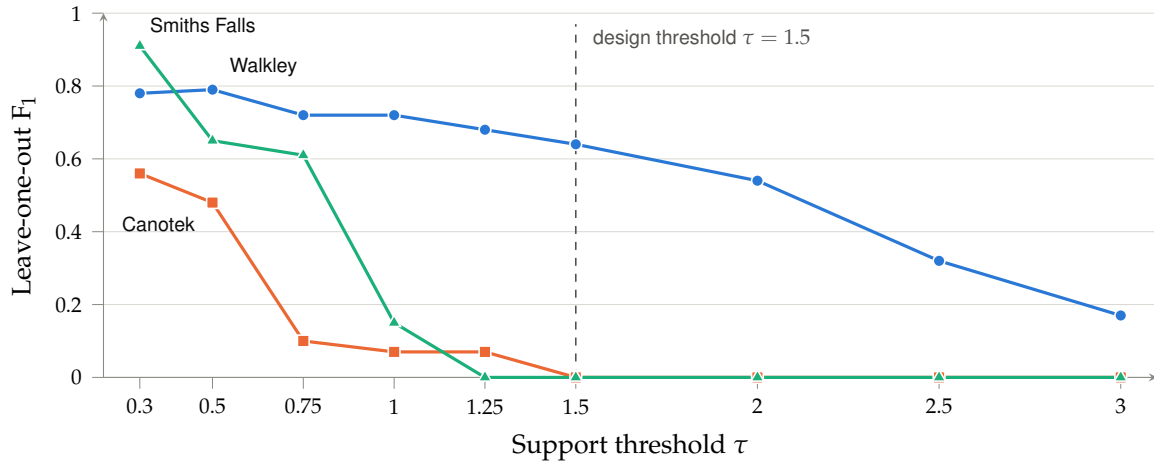


Figure 2: F_1 against threshold. Walkley: 29 traces, 13 authors. Canotek: 14 traces, 8 authors, half-life 3 years. Smiths Falls: 4 usable traces at the default floor, validated with the segment floor lowered to one. The optimum lies at the bottom of the grid at two of three centres, so a lower τ was never tested. Canotek and Smiths Falls overlap at zero above $\tau = 1.25$. The full counts are in Table 8.

are summed over all i (micro-averaging), and precision, recall and $F_1 = 2tp / (2tp + fp + fn)$ are reported. The false-positive scope follows the implementation, and the logged counts are used unchanged. Two variants appear in Section 6. *Pooled* builds one consensus per centre. *Per-family* runs leave-one-out within each route family and pools the family counts.

Eligibility. A trace enters validation only if it contributes at least two distinct segments (lowered to one at Smiths Falls, Section 6.5), and a centre needs at least five usable traces. These floors matter in practice. A new single-segment trace can be real, correctly built and still invisible to the metric. One Canotek addition, a clean account whose two mentions of Montreal Road collapse to a single segment, changed nothing for exactly this reason.

What leave-one-out measures. It asks whether one source’s segments are predicted by the *other* sources. That measures reproducibility, meaning agreement among independent accounts. It is not correctness against the real route. A bias that every source shares, such as cameras reading street signs the car drives past, inflates agreement and goes undetected. Errors that are independent across sources pull agreement down, while shared biases push it up, so every F_1 in this report should be read as agreement, not accuracy. Section 9 describes the reference that now exists.

6 Results

6.1 The design-time threshold was wrong at every centre

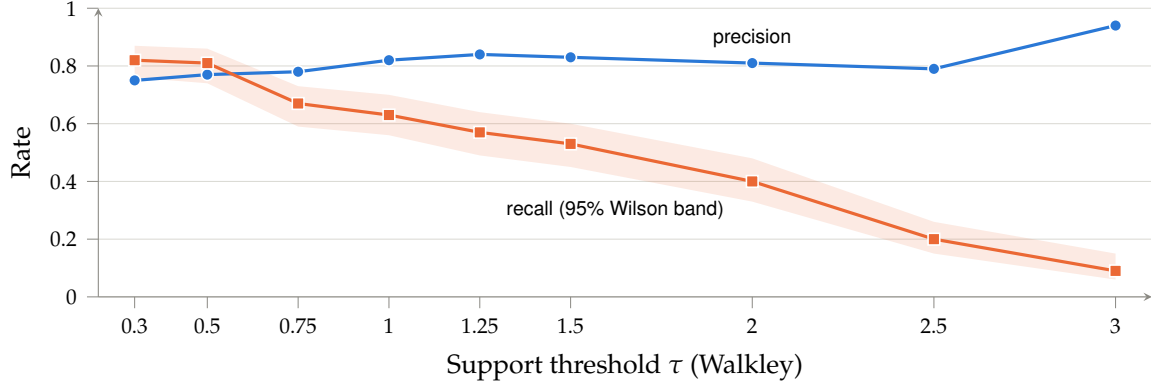
The design fixed a publish threshold of $\tau = 1.5$ by intuition. Figure 2 shows the leave-one-out sweep at each centre, and Table 3 gives the two operating points side by side.

At Walkley, raising τ from 0.5 to 1.5 gives up 48 correct segments to avoid 22 errors, 2.2 lost for every error avoided. Between 1.25 and 1.5 the trade is strictly negative: false positives stay at 19 while seven more true positives are lost. At Canotek and Smiths Falls no segment clears 1.5 at all.

Two caveats apply. First, the best threshold was chosen on the same leave-one-out results it is

Table 3: Operating points. F_1 at the best tested threshold against the design threshold.

	Walkley	Canotek	Smiths Falls
Best tested τ	0.50	0.30	0.30
F_1 at best τ	0.79	0.56	0.91
F_1 at $\tau = 1.5$	0.64	0.00	0.00

**Figure 3:** Walkley precision and recall against threshold. Recall falls steadily while precision stays flat, which is the signature of systematic rather than random false positives. The 0.94 at $\tau = 3$ rests on 17 predictions.

reported with, so the “best” F_1 is optimistically biased. Selecting across centres would be cleaner (Section 8). Second, the grid starts at 0.30, and at two centres the optimum sits on that edge. The conclusion that 1.5 is far too high is robust. The claim that 0.30 is optimal is not.

A precision ceiling. Across $\tau \in [0.3, 2.5]$ at Walkley, precision stays between 0.75 and 0.84 (Figure 3). If false positives were random noise, raising the bar would remove them. They persist, so roughly one prediction in five is wrong for a systematic reason that no threshold can reach. Two causes were identified: cross-route contamination (Section 6.3) and drive-past segments (Section 6.4).

6.2 The weighting did not beat a mention count

Walkley. The weighted estimator and raw frequency both score F_1 0.79 (Table 4). Their posterior difference runs from -0.06 to $+0.07$, and the probability that weighting is better is 0.52. The two differ in operating point, not accuracy. Frequency finds 8 more correct segments and makes 13 more errors. Both beat random decisively, so the method as a whole works. What is unjustified is the sophistication inside it.

Why would deduplication not help? The metric asks whether a segment is predicted, not how strongly. A segment whose support is inflated fourfold by one prolific author is still predicted. Deduplication corrects the *meaning* of the score. It showed up directly in the counts: baycrest $\times$ walkley had 18 mentions but four authors, so counting mentions overstates corroboration $4.5\times$. But it does not change the *ranking* enough to move a threshold decision, and only ranking is being measured.

Canotek. Here weighting was clearly worse: 0.56 against 0.77, a posterior difference of -0.36 to -0.06 under independence. The logged per-trace weights (Table 9) show two mechanisms.

Table 4: Weighted consensus against baselines, leave-one-out, pooled. The brackets give 95% Wilson intervals (precision, recall) and a Jeffreys–Dirichlet posterior interval (F_1), all assuming independent segments. Section 7 relaxes that assumption.

Centre	Method	tp	fp	fn	Precision	Recall	F_1
Walkley $\tau = 0.5$, 29 tr., 13 auth.	Weighted	138	41	33	0.77 [.70,.83]	0.81 [.74,.86]	0.79 [.74,.83]
	Frequency	146	54	25	0.73 [.66,.79]	0.85 [.79,.90]	0.79 [.74,.83]
	Random	58	19	113	0.75 [.65,.84]	0.34 [.27,.41]	0.47 [.39,.54]
Canotek $\tau = 0.3$, $h = 3$, 14 tr., 8 auth.	Weighted	26	10	31	0.72 [.56,.84]	0.46 [.33,.58]	0.56 [.43,.67]
	Frequency	42	10	15	0.81 [.68,.89]	0.74 [.61,.83]	0.77 [.67,.84]
	Random	13	4	44	0.76 [.53,.90]	0.23 [.14,.35]	0.35 [.22,.49]

1. **Age decay dominated.** Ten of 18 traces fell below $\tau = 0.30$ on their own weight. Every 2021 trace landed at $w_{\text{age}} \approx 0.29$ – 0.31 regardless of source, so even video (0.9) was pulled down to about 0.27. A segment supported by one such trace can never be predicted. Author deduplication cannot be responsible: it takes a maximum and never sums, so it cannot push support below any single trace’s weight.
2. **Route imbalance.** Two traces well above threshold (combined weights 0.53 and 0.54) were still mostly unpredicted. Both followed a second, thinner route that other traces rarely corroborate.

The two fixes point in different directions. Adding real traces that corroborate the thin route raised F_1 from 0.56 to 0.63 with no parameter change. Moving the half-life from 3 to 5 years then raised it to 0.77. After that change the weighted and frequency results were *identical*, 44/10/16 on every count. A final reading of the whole 53-thread Reddit backlog added one trace (F_1 0.78 against 0.79).

The identical counts raised a question: had the weighting simply collapsed into counting? A direct comparison of predicted sets says no. On the full corpus, weighting predicts 32 segments and frequency 37, and the 32 are a strict subset of the 37. All five extra segments come from a single older text trace whose combined weight, 0.299, sits just below $\tau = 0.30$. The weighting is doing one small, defensible thing: declining to publish segments that rest on a single near-threshold source.

Two limits on this conclusion should be stated. The half-life was tuned on the Canotek data it is evaluated on. The only guard against overfitting is that Walkley’s numbers did not move under the change, byte for byte, and that is weak evidence. And one trace is the whole difference between the methods, which is the sample size at which such a statement means least.

6.3 Pooling distinct routes, and the instability of families

Class is recoverable from geometry. On the first 17 Walkley traces, HDBSCAN put all 14 G2 traces in one cluster and the G traces in another (two together, one as noise) without being told that test class exists. The G routes use arterials (Hunt Club, Bank, Uplands, Airport Parkway) that no G2 route touches. This supports checking collector class labels against geometry rather than trusting them.

Pooling across routes loses accuracy. Every one of the worst-predicted Walkley traces followed a minority route. Pooled consensus answers a question nobody asked, namely what the average of several different routes looks like. Splitting into families before leave-one-out raised F_1 from 0.79 to

Table 5: Walkley leave-one-out by route family, $\tau = 0.5$, 30 traces, 14 authors (2026-09-04). Per-family pooling uses only the 18 clustered traces.

Family	Traces	Authors	Precision	Recall	F_1	Character
1	11	3	0.95	0.95	0.95	G2, residential loops
2	4	4	0.85	0.79	0.81	G, Albion / Uplands / Lester
3	3	3	0.69	0.65	0.67	G, Airport Pkwy / Hunt Club
Not in a family	12					8 labelled noise, 4 not evaluated
Per-family, pooled			0.88	0.85	0.86	
All traces, pooled			0.75	0.82	0.78	

0.88 on 29 traces, and from 0.78 to 0.86 on 30 traces (Table 5).

But the family numbers are fragile. Four observations limit how far Table 5 can be pushed.

- **Self-similarity.** Family 1’s 0.95 comes from 11 traces written by 3 people. Predicting a held-out trace is easier when a near-copy remains in the training set. Family 2 is the cleanest estimate: four traces, four authors, F_1 0.81.
- **Selective coverage.** The per-family figure is computed on 18 of 30 traces. The 12 excluded are disproportionately text, including all five Reddit accounts of the G route. That tilts the measurement toward video, the source most exposed to drive-past bias.
- **Instability.** The same data clustered as families of 11/4/3 in the morning and 11/3/10 in the evening, after a correct change to name normalisation. After OCR traces were regenerated, the two G families merged and their F_1 fell to 0.67.
- **It did not persist.** Re-measured on 2026-09-10 after the Reddit corpus had grown (30 traces, 27 authors), the families scored 0.87, 0.43 and 0.62. The per-family and pooled figures were then *equal*, at 0.74. The 0.86 was never reproduced, and it should not be quoted as Walkley’s current accuracy.

At Canotek, clustering made things worse. HDBSCAN found one family of 3 traces out of 14, and per-family F_1 fell from 0.56 to 0.43. Family structure needs density that a corpus of this size does not reliably provide.

A related finding came later: clustering on shared streets without class labels merges a G route and a G2 route that share arterials near the centre. Partitioning by confirmed class first, with ambiguous traces attached afterwards, fixed the “G and G2 look identical” symptom on the live map.

6.4 The drive-past bias

Once the Walkley G videos were processed, the best-supported segment in the corpus was colliston $\times$ walkley: 7 authors, 15 video mentions and a weight of 3.96. The author, who knows the area, established from the frame in Figure 4 that the route never turns onto Colliston. It passes straight through a signalised intersection whose overhead signs name both cross-streets.

Table 6 shows why this is structural and not a bug. Each filter passes the segment for a valid reason, and a closed-walk constraint, proposed as a fix, fails too: the route genuinely visits that node.



Figure 4: Dashcam frame from a Walkley G video. Overhead signs name both Colliston (left mast) and Albion N (right). The car goes straight through. OCR reads both correctly, the junction is real, and the sign is visible for a normal approach, so every video-side filter accepts colliston $\times$ walkley as a segment.

Table 6: Every video-side defence passes the drive-past segment.

Filter	Why colliston $\times$ walkley passes
Junction validation	The junction is real, and the route traverses it
Duration filter	The sign is visible for a normal approach
OCR / fuzzy-match check	The read is exact; confirmed by exact-match OCR
Support threshold	7 authors and 15 video mentions clear any realistic τ
Closed-walk constraint	The segment is incident to a node on the route

A camera records what it passes; a person records what they did. Five independent Reddit authors wrote out the G route turn by turn, and none named Colliston. That asymmetry is the only signal found. It suggests a narrow rule: where two segments share a junction and one has text corroboration while the other has only video mentions, flag the video-only segment. The rule is implemented and currently fires zero times, correctly. Only one text trace in the Walkley corpus describes that junction at all, so there is no evidence of a person present at it. The ground truth enters instead through the corrections mechanism, which flags segments and never edits traces.

The bias has a measurable cost that grows as OCR improves. Regenerating the G traces with a better gazetteer read *more* signs, including more signs the car passes, and the G family's F_1 fell from 0.81 to 0.67. Until drive-past segments can be separated, better reading makes the G route estimate worse.

6.5 Smiths Falls: a small-sample result

Smiths Falls had four usable traces when first validated, below the five-trace minimum at the default two-segment floor. Lowering the floor to one segment (both affected traces had been checked against real junctions) and sweeping τ gave F_1 0.91 at $\tau = 0.30$ (21/0/4). The Jeffreys posterior interval on that figure is 0.79–0.97. As more real traces were added, it moved to 0.89

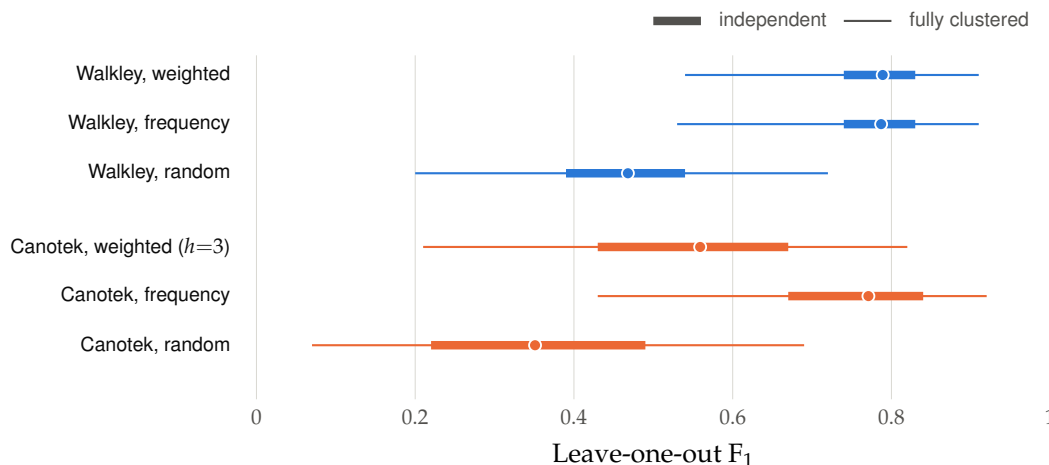


Figure 5: 95% intervals on F_1 under two assumptions. Thick bars treat segments as independent. Thin bars shrink the effective sample to the number of authors (13 at Walkley, 8 at Canotek). The distance between the two is the price of not knowing the within-author correlation.

and then 0.78. That shift is what a small, still-growing corpus should do, and it is why a single number for this centre would mislead. The more durable result is qualitative. The G2 route is fully double-corroborated: a caption-bearing video and an independent Reddit account agree on all 10 of its segments. It is also the one place where the burned-in captions of a single video became a trace.

7 Uncertainty under dependence

Every interval in Section 6 treats segments as independent draws. They are not. Segments cluster within traces, and traces cluster within authors. One Walkley author contributed four traces, and at Canotek three uploads came from one channel. Author deduplication handles this inside the estimator. The evaluation's standard errors do not.

The right tool is a cluster bootstrap that resamples authors, reruns leave-one-out and recomputes F_1 . That needs per-trace outputs, and the project logs record only aggregate counts. As a transparent substitute, two bounds are reported.

- **Independent.** A Jeffreys Dirichlet posterior on the (tp, fp, fn) shares, mapped through $F_1 = 2p_{tp} / (2p_{tp} + p_{fp} + p_{fn})$ with 200,000 draws.
- **Fully clustered.** The same posterior with the counts scaled so the effective sample size equals the number of authors. This is the design effect when segments within an author are perfectly correlated, the worst case.

The truth lies between the two.

Figure 5 and Table 7 show the result. Under independence, the Walkley F_1 is pinned to within about ± 0.05 . Under full clustering the same number is compatible with anything from 0.54 to 0.91. Three conclusions survive the wider bound.

1. At Walkley, the weighted and frequency methods cannot be told apart under either assumption.
2. At Walkley, both beat random under independence. Under full clustering their intervals still

Table 7: Interval width under the two dependence assumptions.

Centre	Method	F_1	95% interval	
			Independent	Fully clustered
Walkley (13 authors)	Weighted	0.79	0.74–0.83	0.54–0.91
	Frequency	0.79	0.74–0.83	0.53–0.91
	Random	0.47	0.39–0.54	0.20–0.72
Canotek (8 authors)	Weighted, $h = 3$	0.56	0.43–0.67	0.21–0.82
	Frequency	0.77	0.67–0.84	0.43–0.92
	Random	0.35	0.22–0.49	0.07–0.69

overlap random’s upper tail, so even that claim rests on the segments being at least partly independent.

3. At Canotek, the 0.21 gap between weighted and frequency at $h = 3$ is clear under independence (-0.36 to -0.06) but would not be clear under full clustering. With eight authors, differences of this size sit at the edge of what the data can resolve.

The practical reading: **the number of independent authors, not the number of segments, bounds what this project can learn.** Walkley’s 171 evaluated segments come from 13 people. Adding traces from people already in the corpus narrows the independent interval without narrowing the true uncertainty. Adding new authors narrows both.

8 Threats to validity

Construct validity: agreement is not accuracy. Leave-one-out rewards consistency among sources. A shared bias, such as drive-past segments, relayed copies of one route list, or a gazetteer that systematically mangles one street name, looks like consensus and scores well. The one direct test of correctness available during development was local knowledge applied to specific segments, and it overturned the highest-support segment in the corpus.

Non-independence. The dependence analysed in Section 7 also biases the point estimates, not just their spread. One Walkley author posted their own route and then, in a reply, relayed three routes they had taken from YouTube. The two accounts differed by one residential leg. That difference was first logged as an error in the relayed copy, and later shown by three independent videos to be a genuine route variant: the correction itself had been recorded too broadly. An early claim of “three independent authors” agreeing on a sequence turned out to be two; the salted author hash caught it. The reliability model still does not distinguish firsthand from relayed accounts, mock tests from real tests, or route-guide overlays from dashcam recordings. The same distinction surfaced three separate times.

Selection. The corpus is whatever people chose to publish. Reddit route write-ups plausibly skew toward people who passed, or who were answering a thread that asked for routes; this was not measured. Instructor “route guide” videos may show roads a real test does not use. One “Hawthorne” mock-test video shared no segment with 14 other G traces. The per-family validation, meanwhile, systematically excluded text traces (Section 6.3).

Temporal drift. The corpus spans 2017 to 2026, and routes change. The age decay encodes that belief but has never been tested against it. The half-life moved from 3 to 5 years because of a leave-one-out score, not because of evidence about how fast routes actually change. A moderator’s claim that G2 routes are stable while G routes are not is plausible, came from a single source and is unverified. If true, one decay curve for both classes is wrong.

Researcher degrees of freedom. Thresholds and the half-life were tuned on the data they are reported with, so the headline figures are optimistic. A cleaner protocol would choose τ and h on some centres and report on another. With three centres, that becomes leave-one-centre-out. Several other parameters were set by reasoning and never measured: the fuzzy-match cutoffs, the 15s duration filter, the ± 2 junction window, the source weights and the 0.15 age floor. The one parameter that was checked, $\tau = 1.5$, turned out to be off by a factor of three to five.

Instrument failures. At least five times during development, the pipeline produced a confident, wrong result with no internal sign of failure. A route omitted an entire crescent while reporting 10 of 10 junctions resolved. Two separate G routes were welded into one plausible 15.1 km path. A street-detection rate of 88% came from matching “looking good” and “green light” against real street names. An OCR run reported zero text across 1,095 frames because every frame had thrown a swallowed exception. And 0 of 112 transcripts came back because an API interface had changed. Each was caught by a person noticing something implausible, never by the system. The standing lesson, applied throughout the numbers here: *validate the instrument before the hypothesis*. Appendix D lists the main figures in the project log that were later corrected.

9 From estimator to product

The site went live at ontariodrivetestmap.fyi with a FastAPI backend and a Postgres database. Getting from a validated segment set to a drawable route taught three further lessons about what the estimator can and cannot supply.

Segments are not routes. The consensus says *which* segments are well supported. It does not say how many distinct routes exist or in what order a route visits them. Overlap-based clustering cannot tell three short routes apart from one long route seen in fragments. One attempt fixed each centre’s route count by hand and let a travelling-salesman tour through the consensus junctions supply the order. It produced connected routes, and a Walkley G loop of 36 km, longer than a real test.

Inference must be visible as inference. Bridging gaps between confirmed segments of the same family with road-snapped paths makes a route readable. It also creates geometry that no source described. An earlier attempt drew straight lines between confirmed points and cut visibly through buildings. It was reverted. A road-snapped fabrication is worse, because it looks right. The rules that survived are these. Bridges connect only points already in the same trace-verified family. They follow a minimum spanning tree with a distance cap. They carry a flag in the data. And they are drawn in a notation that never borrows the confirmed style. The live map now uses one notation everywhere: *Confirmed* (solid), *Inferred* (dashed), *Unrouted gap* (dotted).

The routes on the map are hand-transcribed. Since 2026-09-18, the 13 route lines on the live map are the author’s own street-by-street transcriptions of test videos, resolved against the same junction index and routed through OSRM. That covers three G and three G2 routes at Walkley, two G and three G2 at Canotek, and one of each at Smiths Falls. The transcription overturned the automated route counts at three of six centre-class pairs: Walkley G has three routes, not one, and Canotek has two G routes and three G2, not one of each. Resolving the transcripts also found two real geometric traps. One junction pair meets twice about 200 m apart around a crescent, and a transcribed “Regional Rd” turned out to be Jasper Road, which becomes Jasper Avenue at a three-way junction. The consensus segments remain on the map as the evidence layer. Of 126 transcribed turns, 92 were anchored to a real junction within 40 m of the drawn route, in driving order on all 13 routes.

What this means for the statistics. The hand-transcribed routes are the closest thing the project has to ground truth, and they are *not* used in any result above. They are an obvious external reference for the automated estimator, with one important caveat: they were transcribed from the same kind of video that feeds the OCR traces, possibly including some of the same videos. Agreement with them is therefore not fully independent of the video evidence. The drive-past error, at least, should not carry over: a person watching the car go straight does not write down the cross-street.

10 Future work

In rough order of value per unit of effort:

1. **External validation.** Score the automated consensus against the hand-transcribed routes, per centre and class, with a cluster bootstrap over authors. This turns agreement into something closer to accuracy, and it would test whether drive-past segments are the main source of false positives.
2. **Leave-one-centre-out parameter selection.** Choose τ and h without looking at the centre being scored, and extend the grid below 0.30.
3. **A per-segment posterior.** Model route membership as a beta-binomial over independent authors, so that two segments with the same author count but different strength of evidence get different credible intervals. This is what the map’s line styling tries to express with three categories.
4. **Learned source reliability.** A Dawid–Skene model would estimate each source type’s reliability from its agreement with consensus instead of asserting it. It could also learn directly that video is complete but over-inclusive at signalised junctions, while text is incomplete but specific.
5. **A source_kind field.** Mark firsthand, relayed, mock and guide sources at collection, so the distinction that surfaced three times becomes data rather than a filter.

11 Conclusion

Crowd-reported road-test routes can be reconstructed. The central premise held repeatedly on real data: strangers years apart, in different media, describe the same turns. What the project measured is how that reconstruction fails. The parameter chosen by intuition was wrong at every

centre where it was checked. The weighting scheme, with every component argued for from a real observation, did not beat counting until it was retuned on the data it is scored on. Route families improved agreement until the corpus grew. And the most complete source, video, carries a bias that only the least complete source, text, can expose.

The number of independent people who took the time to write down or record a route bounds what any of this can know, and at every centre that number is small. The honest output of this project is therefore not a route but a route with its uncertainty attached, and a record of every place the method was wrong before it was right.

A Threshold sweeps

Table 8: Leave-one-out threshold sweeps, weighted consensus, pooled. Counts as logged. Precision is undefined where nothing is predicted.

τ	Walkley				Canotek ($h = 3$)				Smiths Falls			
	tp	fp	fn	F_1	tp	fp	fn	F_1	tp	fp	fn	F_1
0.30	141	48	30	0.78	26	10	31	0.56	21	0	4	0.91
0.50	138	41	33	0.79	20	7	37	0.48	12	0	13	0.65
0.75	114	33	57	0.72	3	0	54	0.10	11	0	14	0.61
1.00	108	23	63	0.72	2	0	55	0.07	2	0	23	0.15
1.25	97	19	74	0.68	2	0	55	0.07	0	0	25	0.00
1.50	90	19	81	0.64	0	0	57	0.00	0	0	25	0.00
2.00	69	16	102	0.54	0	0	57	0.00	0	0	25	0.00
2.50	34	9	137	0.32	0	0	57	0.00	0	0	25	0.00
3.00	16	1	155	0.17	0	0	57	0.00	0	0	25	0.00

The Smiths Falls log reports $\tau \geq 1.25$ as a single row with F_1 0.00. The counts shown there follow from recall 0 on 25 held-out segments.

B Canotek trace weights

Table 9: Combined weight $w = w_{\text{src}} w_{\text{age}}$ for the 18 Canotek traces at $h = 3$ (logged 2026-09-10). Trace identifiers are omitted. Rows below the operating threshold $\tau = 0.30$ are marked *.

Source	Turns	Date	w_{src}	w_{age}	w
Reddit	5	2026-08	0.45	0.98	0.440
Reddit	5	2022-08	0.45	0.39	0.177 *
Reddit	4	2025-08	0.45	0.79	0.355
Reddit	4	2025-08	0.45	0.79	0.355
Reddit	2	2021-08	0.25	0.31	0.078 *
Reddit	3	2021-08	0.45	0.31	0.141 *
Reddit	2	2024-08	0.25	0.63	0.156 *
Reddit	12	2021-08	0.60	0.31	0.188 *
Reddit	8	2022-08	0.60	0.39	0.237 *
Reddit	5	2024-08	0.45	0.63	0.282 *
Reddit	4	2025-08	0.45	0.79	0.355
Video	15	2021-05	0.90	0.29	0.263 *
Video	17	2024-06	0.90	0.59	0.535
Video	3	2021-04	0.90	0.29	0.259 *
Video	4	2025-10	0.90	0.81	0.731
Video	4	2024-06	0.90	0.60	0.539
Video	7	2024-06	0.90	0.59	0.534
Video	15	2021-07	0.90	0.31	0.275 *

C Parameters and how each was set

Table 10: Every tunable in the estimator, and whether it was ever checked against data.

Parameter	Value	Basis
Publish threshold τ	1.5 designed; 0.5 Walkley, 0.30 Canotek and Smiths Falls	Measured (sweep)
Age half-life h	3 years designed; 5 years adopted	Tuned on Canotek
Age floor; undated weight	0.15; 0.6	Reasoned
Source weights	video 0.9; Reddit 0.25 / 0.45 / 0.60	Motivated by completeness, not fitted
Fuzzy match (text)	0.86 local list, 0.88 full gazetteer	Reasoned (asymmetric costs)
Fuzzy match (OCR)	0.85	Reasoned
OCR duration filter	≈ 15 s	Separated perfectly on one video
OCR burst filter	≥ 4 streets per frame	Observed overlays
Junction window	± 2 sightings	Fixed after a real miss
Minimum family size	3 traces	Reasoned
Validation floors	≥ 2 segments per trace; ≥ 5 traces	Reasoned

D Corrections ledger

Figures recorded during the project and later found to be wrong. Every one of them was a plausible number when it was first written down.

Table 11: Corrections ledger.

Recorded	Actually	How it was caught
Three independent Reddit authors agree on a G2 sequence	Two; a post and a reply by the same author	Salted author hash
0 of 112 videos have transcripts	Library interface changed; the error was swallowed	0 of 112 was implausible
88% of transcripts name streets next to turns	Gazetteer false friends (“good”, “green”); 0 traces	Restricting the gazetteer to streets near the centre
OCR finds no text in 1,095 frames	Backend crashed on every frame	“No text at all” is physically impossible for dashcam footage
A route fully resolved, 10 of 10 junctions	One crescent silently collapsed and never driven	Local knowledge, looking at the map
Heron/Sandalwood leg disputed	A real variant, read in three videos	Scope of the correction re-examined
Canotek F_1 is 0.86	That was Walkley’s; Canotek’s first real figure was 0.56	Tracing the number’s source
A reference Smiths Falls video carries one real street read	All hits were business names	Reading the full OCR context
Canotek and Smiths Falls consensus files current	5 days stale; a report-only flag wrote nothing	Migration showed zero rows
Three G/G2 mixed-class routes	Ambiguous traces were voting as a class	Checking one layer further down

E Reproducibility

Every count in this report was copied from the project’s build log and report-material notes, each written when the measurement was made. The intervals, the posterior comparisons and the Fisher test were computed from those counts by a short Python script (`stats.py`), which also asserts that every logged F_1 reproduces from its confusion counts to two decimals, and that the logged age and IDF weights match Equation (2) and the IDF form in Section 4.3. The code is split by design: consensus and geometry are meant to be public, while the acquisition layer stays private, because it holds API credentials and to avoid republishing third-party content. Author identities are stored only as salted hashes.